GREEN SPACE MAPPING IN BIRGUNJ METROPOLITAN CITY, PARSA, NEPAL

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GREEN SPACE MAPPING IN BIRGUNJ METROPOLITAN CITY, PARSA, NEPAL

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Susmita Khadka

LETTER OF ACCEPTANCE

This is to certify that the thesis entitled, "Green Space Mapping in Birgunj metropolitan city, Parsa Nepal" submitted in partial fulfilment for the requirements for the degree of **Bachelor of Science in Forestry** of the undergraduate program, Agriculture and Forestry University, Nepal is a record of original research carried by **Ms. Susmita Khadka, Reg. No. 3-1-132-2021**, under my guidance and supervision and no part of this thesis has been submitted for any other degree or diploma.

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I, Susmita Khadka, hereby declare that this project paper under internship entitled **"GREEN SPACE MAPPING IN BIRGUNJ METROPOLITAN CITY, PARSA, NEPAL"** is based on primary work and all the sources of information used are duly acknowledged. This work has not been submitted to any other university for any academic award.

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ACRONYMS

AGB: Above Ground Biomass

ArcGIS: Arc Geographical Information System

BMC: Birgunj Metropolitan City

CBS: Central Bureau of Statistics

DBH: Diameter at Breast Height

ESA: European Space Agency

GBH: Girth at Breast Height

GIS: Geographical Information System

GPS: Global Positioning System

IVI: Important Value Index

LULC: Land Use Land Cover

MLC: Maximum Likelihood Classification

MPFS: Master Plan for Forestry Sector

NDVI: Normalized Difference Vegetation Index

OBIA: Object Based Image Analysis

RS: Remote Sensing

SDGs: Sustainable development goals

S2A: Sentinel-2A

UGS: Urban Green Space

UN: United Nation

WHO: World Health Organization

Abstract

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As Nepal's primary economic and commercial gateway, Birgunj Metropolitan City is experiencing rapid, often unplanned, urban expansion that threatens its ecological integrity. This research provides a comprehensive assessment of the city's urban green spaces, moving beyond simple mapping to quantify vegetations present over there, their composition and stem biomass estimation. Utilizing high-resolution Sentinel-2A satellite imagery and intensive ground-truthing across 302 sample plots, a supervised classification was performed to map green spaces, while Above Ground Biomass (AGB) was estimated through existing allometric equations. The land use classification, validated with a high Kappa coefficient of 0.871, revealed that Birgunj's landscape is dominated by agricultural land (53.06%), followed by human-settlements (20.53%), public grounds (18.37%) and water bodies (8.04%). Further, ecological assessment of 23 tree species identified a delicate balance between native (52%) and exotic (48%) species. While *Eucalyptus spp.* (Masala) were found to be the largest carbon sinks (averaging 153.73 kg/ha), species such as *Dalbergia sissoo* (Importance Value Index = 53.99) and *Neolamarckia cadamba* (Relative Coverage = 25.86%) were identified as the most ecologically dominant species. Thus, as only one-fourth of open land is under government jurisdiction, maintenance of planned green spacing in Birgunj requires collaboration with private agricultural land holders. Policy instruments such as community-based programs, financial incentives, and awareness initiatives could play a pivotal role in aligning respective public and private land-use practices with broader environmental and urban planning objectives.

Keywords: *Sentinel-2A, Supervised classification, Above Ground Biomass*

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Major Advisor

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शोध सार

नाम: सुस्मिता खड्का दर्ता नम्बर: ३-१-१३२-२०२१

सत्र तथा भर्ना बर्ष: प्रथम,२०२१ उपाधि: वन विज्ञान स्नातक

प्रमुख सल्लाहकार: प्रध्यापक सुमन घिमिरे

नेपालको प्रमुख आर्थिक तथा व्यापारिक प्रवेशद्वारको रूपमा रहेको वीरगञ्ज महानगरपालिका तीव्र र प्राय: अनियोजित सहरी विस्तारको अनुभव गरिरहेको छ जसले यस क्षेत्रको पारिस्थितिकीय अखण्डतलाई खतरामा पारेको छ।यस अनुसन्धानले शहरी हरित क्षेत्रहरूको व्यापक मूल्याङ्कन गरेको छ, जसमा केवल साधारण नक्साङ्कन मात्र नभई त्यहाँ रहेको वनस्पतिहरू, तिनीहरूको संरचना र तना जैवभारको अनुमानसमेत गरिएको छ।उच्च रिजोल्युसनयुक्त सेन्टिनल-२ए उपग्रह चित्र तथा ३०२ नमुना प्लटमा गरिएको गहन स्थल सत्यापनको आधारमा पर्यजवेक्षित वर्गीकरण विधि प्रयोग गरी हरित क्षेत्र नक्साङ्कन गरिएको हो। भूमिस्तरीय जैवभारको गणना विद्यमान एलोमेट्रिक समीकरणहरू प्रयोग गरी गरिएको हो।प्राप्त जमिन उपयोग वर्गीकरण ०.८६१को उच्च कप्पा गुणांकसहित प्रमाणित भएको छ, जसले यसलाई अत्यतै विश्वसनीय बनाउँछ।नतिजाले देखाएको छ कि वीरगञ्जको भुमीमा कृषियोग्य जमिन ५३.०६ प्रतिशत प्रमुख छ भने त्यसपछाडि मानव बस्ती २०.५३ प्रतिशत, सार्वजनिक मैदान १८.३७ प्रतिशत र जलाशय ८.०४ प्रतिशत रहेका छन्।पारिस्थितिकीय मूल्याङ्कनमा २३ प्रजातिका रूखहरू पहिचान भएका छन्, जसमा ५२ प्रतिशत स्वदेशी र ४८ प्रतिशत विदेशी प्रजातिहरू रहेका छन्।मसाला सबैभन्दा ठूलो कार्बन सङ्कका रूपमा औसत १५३.८६ किलोग्राम प्रति हेक्टर पाइयो।त्यस्तै, सिसौ ५३.९९को महत्व सूचकाङ्कका साथ सबैभन्दा प्रभुत्वशाली प्रजाति देखियो भने कदम्ब २५.८६ प्रतिशतको सापेक्ष आच्छदनका साथ प्रमुख प्रजाति पाइयो।खुला जमिनको केवल एक चौथाइ भाग मात्र सरकारी स्वामित्वमा भएकले वीरगञ्जमा योजनाबद्ध हरित क्षेत्रको संरक्षण तथा विस्तारका लागि निजी कृषियोग्य जमिनधनीहरूसँग सहकार्य आवश्यक देखिन्छ।यसका लागि सामुदायिक कार्यक्रमहरु, आर्थिक प्रोत्साहनहरू तथा जनचेतना अभिवृद्धि जस्ता नीतिगत उपकरणहरूले सार्वजनिक र निजी जमिन प्रयोगलाई दीर्घकालीन वातावरणीय तथा योजना उद्देश्यहरूसँग संरेखित गर्न महत्वपूर्ण योगदान पुऱ्याउन सक्छन्।

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प्रध्यापक सुमन घिमिरे सुस्मिता खड्का

प्रमुख सल्लाहकार लेखक

Chapter One: INTRODUCTION

1.1 Background

Urbanization has become one of the most defining trends of the 21st century, fundamentally transforming human settlements, economies, and the environment. Driven primarily by the pursuit of better economic prospects, education, healthcare, and overall quality of life, rural-to-urban migration has significantly contributed to the rise in urban population globally. As of 2024, more than 4.52 billion people i.e. over 57% of the global population reside in urban areas (Ritchie et al., 2024). This trend is expected to continue, with projections indicating that by 2050, nearly 70% of the world's population will be residing urban (UN, Department of Economic and Social Affairs, Population Division, 2019). While urbanization can serve as a catalyst for innovation, economic development, and modernization, it also presents profound challenges to environmental sustainability, urban planning, and human well-being. In many developing countries, urbanization is particularly rapid and often unplanned or poorly managed, leading to a range of environmental and socio-economic issues (Angel et al., 2011; Satterthwaite et al., 2020). The expansion and densification of urban populations have escalated the demand for land, water, energy, housing, transportation, and other infrastructure, placing enormous pressure on already fragile urban ecosystems (Adedeji, 2023).

As cities expand, they tend to consume vast tracts of agricultural land, wetlands, forests, and other natural ecosystems, thereby disrupting local biodiversity and ecological balance. This uncontrolled growth typically results in significant land use and land cover (LULC) changes, including the conversion of green spaces, farmlands, and natural vegetation into impervious surfaces such as concrete buildings, roads, and pavements. The transformation of landscapes due to urbanization often leads to the degradation or complete loss of urban green spaces, which are critical for environmental functions like a purification, temperature regulation, and stormwater absorption, and human well-being (Seto, Güneralp, & Hutyra, 2012; UN-Habitat, 2020; Kabisch et al., 2015). According to UN-Habitat (2020), urban growth in many regions, especially in South Asia and Sub-Saharan Africa, occurs without sufficient spatial planning, resulting in informal settlements, urban sprawl, and limited access to green and open areas. These changes compromise essential ecosystem services such as air and water purification, climate regulation, carbon sequestration, and urban temperature moderation. Additionally, the lack of green infrastructure exacerbates urban heat island effects, flood risks, and air pollution, while also reducing opportunities for recreation and social cohesion among city residents particularly absence or uneven distribution of green areas in cities can contribute to social exclusion, poor mental health, and limited physical activity, particularly in marginalized communities (Gill et al., 2007; Bowler et al., 2010; Jennings et al., 2017; WHO, 2016).

Urban green space (UGS) are those areas within an urban environment that are free from buildings and roads, encompassing a wide variety of land types such as parks, playground, open fields, streets tree corridors, agricultural plots and other public lands. These spaces play a crucial role in promoting ecological balance, providing recreational opportunities and enhancing the quality of life in cities (Barbet et al.,2016). UGS can be broadly categorized into public and private spaces, depending on their ownership, access, and use. Public green spaces include areas like parks, gardens, recreational fields, and green corridors along streets or rivers that are open and accessible to all residents, contributing significantly to social inclusion, recreation, and community health. In contrast, private green spaces are typically found within the premises of homes, institutions, or commercial areas such as private gardens, courtyards, or green rooftops and are usually accessible only to specific individuals or groups (Dai, 2011). While both types play a role in urban ecology, public green spaces are particularly crucial from a policy and planning perspective due to their broader societal and environmental benefits. In the context of this research, the term UGS refers to both publicly and privately accessible green spaces, as both types contribute to vegetation biomass storage and offer ecological and social functions. Private green spaces, although not universally accessible, play a crucial role in carbon sequestration, biodiversity conservation, and can serve as emergency escape zones during natural disasters or pandemics. This multifunctionality of green spaces was highlighted prominently during the COVID-19 pandemic, where the presence of green environments regardless of public or private designation proved essential for mental well-being, physical activity, and stress relief amid lockdowns and mobility restrictions (Astell-Burt et al., 2014; WHO, 2016). In the case of Nepal, the importance of green space was further emphasized following the 2015 Gorkha Earthquake (2072 BS), when urban residents increasingly recognized the need for accessible open areas not only for daily use but also as safe evacuation zones during emergencies. In response, government agencies and urban planners began to pay greater attention to the integration of green infrastructure in urban design and disaster-resilient planning (MoUD, 2016). Maintaining green space in urban area provides environmental, economic, social and psychological well-beings to humankind (Haq.,2011).

The World Health Organization (WHO, 2010) recommends a minimum of 9 sq. M of green open space per city dwellers to support mental well-being, physical activity, and social interaction. This benchmark reflects growing recognition of the role that green spaces play in promoting urban health and environmental sustainability. However, despite this widely accepted guideline, many cities across the globe particularly in the Global South fail to meet this minimum standard. Rapid urbanization, unregulated land conversion, limited governance capacity, and fragmented urban planning systems often result in an unequal distribution of green spaces, with marginalized and low-income communities suffering the greatest deficits (World Bank, 2021). In response to these challenges, the Sustainable Development Goals (SDGs) specifically Goal 11 (Sustainable Cities and Communities) underscore the importance of inclusive, safe, resilient, and sustainable urban environments. One of the key targets under SDG 11 is to "provide universal access to safe, inclusive and accessible, green and public spaces, in particular for women and children, older persons and

persons with disabilities." This highlights the crucial need to ensure that green infrastructure is not only present but also equitably distributed and accessible to vulnerable populations who often lack the means or opportunity to engage with natural environments within the urban fabric. As urban populations grow and climate change pressures intensify, green infrastructure has emerged as a fundamental element of resilient and sustainable city design. According to Tzoulas et al. (2007), green infrastructure including parks, urban forests, green corridors, gardens, and vegetated rooftops provides multiple ecosystem services such as air purification, urban temperature regulation, stormwater management, noise reduction, and biodiversity conservation. It also plays a crucial socio-cultural role by fostering community cohesion, encouraging physical activity, and offering spaces for recreation and psychological restoration.

In this study, the scope expands beyond contributions of UGS by incorporating the concept of above ground biomass (AGB), i.e. total mass located above forest vegetation (Pan et al., 2011) is a crucial to assess the ecological benefit of urban trees, including their carbon sequestration capacity, their contribution to air quality improvement, and their role in mitigation the urban heat island effect, which is crucial for sustainable urban planning and management, especially in densely populated cities (Lin et al., 2022). Urban trees are planted, grown and maintained in open space compared to natural forest (Elmqvist et al., 2015). Trees in Urban Green Spaces (UGS) offer a multitude of ecosystem services that significantly enhance the quality of urban life. They contribute to microclimate regulation by lowering surface temperatures and providing shade and improve aesthetic appeal while reducing urban noise levels. Urban trees act as natural air purifiers, filtering airborne pollutants and sequestering particulate matter, in addition to providing habitat and shelter for diverse bird species and urban wildlife (Nagendra & Gopal, 2010; Nowak et al., 2013). Among these benefits, carbon sequestration through photosynthesis has garnered increasing attention. Urban vegetation captures atmospheric carbon dioxide (CO_2) and stores it in aboveground biomass (AGB) and belowground biomass, playing a vital role in mitigating climate change at the local scale. Despite their ecological importance, urban forests and green spaces have received limited attention in global carbon research, which traditionally focuses on natural forests, afforestation, and reforestation projects outside of urban areas (Myeong et al., 2006; Stoffberg et al., 2010; Zhao et al., 2010; Liu & Li, 2012). These natural ecosystems have been prioritized under UNFCCC's climate change mitigation mechanisms, such as the Clean Development Mechanism (CDM) and Reducing Emissions from Deforestation and Forest Degradation (REDD/REDD+), given their formal recognition in international environmental frameworks (Rosendal & Schei, 2014). However, recent studies in both developed and developing countries have revealed that urban green spaces can be rich in biodiversity and store significant quantities of carbon, particularly in urban trees and soil organic carbon pools (Nowak et al., 2009, 2010, 2011, 2012, 2013). Biomass estimation reflects the carbon storage potential of green spaces, which is critical in the context of climate change mitigation, especially for countries like Nepal that are vulnerable to environmental risks but rich in biodiversity. AGB not only serves as an indicator of vegetation health and density but also provides insights into carbon sequestration capacities in urban settings.

Despite the relevance, there is a lack of localized studies that link UGS mapping with AGB estimation using remote sensing (RS) and GIS techniques in Nepal.

For urban planners and environmental managers in cities, enhancing and improvement of urban green space (UGS) is becoming increasingly critical, especially in the context of rapid urban expansion (van den Bosch & Ode Sang, 2017). Planners are now advocating for the establishment of standards related to the spatial distribution, accessibility, and quality of green spaces to ensure they serve the urban population effectively. Moreover, the adoption of participatory planning approaches is emphasized to tailor UGS development to the needs of different community groups, including the elderly, children, low-income households, and marginalized populations (Mwendwa & Giliba, 2012). To address these issues, researchers and planners have increasingly turned to geospatial and statistical tools. Geographical Information Systems (GIS) are widely used to map, quantify, and monitor green spaces, enabling better planning and visualization. However, only a few localized studies have attempted to link Remote sensing and GIS-based spatial analyses with AGB.

Remote sensing involves acquiring data about the Earth's surface without direct contact, typically using satellite or aerial platforms. It enables the monitoring of land cover changes over time by capturing reflected radiation in various spectral bands. Different land cover types such as vegetation, water bodies, and built-up areas reflect and absorb electromagnetic radiation differently, making them distinguishable using RS data (Campbell & Wynne, 2011). Remote sensing plays a crucial role by providing broad spatial coverage, high temporal resolution, and the ability to automate the process of classification and monitoring as well as quantify green spaces across urban landscapes with reliable spectral information (Xie et al., 2008). Similarly, advancement in optical remote sensing with large scale availability of good quality high resolution satellite data. In order to handle the complexity of the urban environment and provides both spectral and spatial information which is effective for high resolution remote sensing data (Blaschke et al., 2014). For green space studies, vegetation indices such as the Normalized Difference Vegetation Index (NDVI) are commonly used. NDVI is calculated from the red and near-infrared bands and provides an effective measure of vegetation greenness and health (Rouse et al., 1973; Tucker, 1979). High NDVI values generally correspond to dense and healthy vegetation, making it an ideal indicator for mapping urban green spaces and estimating vegetation biomass (Xie, Sha, & Yu, 2008; Pettorelli et al., 2005). In this study, satellite imagery Sentinel-2A is utilized to classify and map the spatial extent of green areas in Birgunj. These high-resolution images offer spatial resolution, enabling seasonal and annual monitoring of green space trends. (Drusch et al., 2012; European Space Agency [ESA], 2023). GIS complements remote sensing by offering advanced tools for spatial data management, visualization, and analysis (Burrough & McDonnell, 1998; Longley et al., 2015). It facilitates the integration of various data layers such as land use, road networks, administrative boundaries, and vegetation indices, allowing planners to analyse green space in relation to human activities and urban infrastructure (Weng, 2010). Through supervised classification techniques in GIS, satellite imagery can be categorized into distinct land use/land cover classes such as built-up areas,

water bodies, agricultural lands, and ground. Supervised classification methods (e.g., Maximum Likelihood, Support Vector Machine) use training samples to guide the classification process, improving the accuracy of green space mapping (Congalton & Green, 2009; Pal & Mather, 2005). Moreover, GIS enables the calculation of proximity, fragmentation, connectivity, and accessibility of UGS, which are crucial parameters for urban ecological planning and the creation of high-resolution maps for visualizing spatial patterns of green infrastructure across administrative units, such as wards or neighbourhoods (Herold et al., 2005; McGarigal et al., 2012; Tomlinson et al., 2011; Kabisch & Haase, 2013).

1.2 Rationale of study

Rapid urbanization has emerged as a significant driver of land-use and land-cover changes across the globe, particularly in developing countries such as Nepal. This transformation often results in the degradation of natural ecosystems, including the reduction and fragmentation of urban green spaces (UN-Habitat, 2020; Ritchie et al., 2024). Urban green spaces (UGS) play a crucial role in providing essential ecosystem services such as carbon sequestration, temperature regulation, biodiversity conservation, and the enhancement of mental and physical health among city dwellers (Nowak, Crane, & Stevens, 2006; Tzoulas et al., 2007; Astell-Burt, Feng, & Kolt, 2014). However, the unplanned expansion of cities has led to the reduction in both the quantity and quality of these spaces (Adedeji, 2023).

To address these environmental challenges, it is necessary to assess and monitor UGS using geospatial technologies. Remote sensing (RS) and Geographic Information Systems (GIS) provide powerful tools for identifying and analysing green spaces at various spatial and temporal scales (Xie, Sha, & Yu, 2008). Estimating Above Ground Biomass (AGB) in urban areas helps quantify carbon storage potential, which is essential for climate mitigation and for understanding urban ecological dynamics (Myeong, Nowak, & Duggin, 2006; Liu & Li, 2012).

Moreover, identifying and assessing native versus exotic species composition within UGS is crucial for promoting biodiversity and ensuring the resilience of urban ecosystems (Kendal, Williams, & Williams, 2012). In Nepal, the importance of green spaces was particularly highlighted after the 2015 Gorkha earthquake and during the COVID-19 pandemic, where green spaces played a vital role in supporting public well-being and emergency resilience (Webster & Sanderson, 2013; Markevych et al., 2017).

Given this context, the present study focuses on mapping urban green spaces and estimating above ground biomass in Birgunj Metropolitan City using satellite imagery and GIS. This research aims to contribute to the development of sustainable urban environments by providing scientific evidence to support green infrastructure planning, carbon accounting, and biodiversity conservation in rapidly urbanizing regions of Nepal. Thus, this study is

purposed with the overarching goal of promoting sustainable and environmentally conscious urban development.

1.3 Objectives

1.3.1 General Objectives

The overall objective of this study is to imply remote sensing technology for mapping green space in Birgunj metropolitan city.

1.3.2 Specific Objectives

- To quantify the urban green space in the Birgunj metropolitan city
- To determine above ground biomass of tree species of prevailing urban green space
- To identify native/exotic species composition in urban green spaces

1.4 Research Question

- How the green space in Birgunj city is distributed?
- What kind of tree species are dominating the green space?
- How much above ground biomass is stored in Birganj's urban green spaces?

Chapter Two: LITERATURE REVIEW

2.1 Evolution of Green space and mapping technique

Before the 19th century, green spaces in urban areas primarily consisted of private gardens owned by the elite, sacred religious grounds, and utilitarian commons. These spaces were not intended for public use and offered little community benefit, reflecting an urban ideology that viewed nature as separate from the functional and social fabric of cities (Mumford, 1961). This perspective began to shift with the onset of the Industrial Revolution, which brought rapid urbanization, overcrowding, pollution, and frequent public health crises in cities like London, New York, and Manchester. In response, the Sanitary Reform Movement emerged, advocating for the integration of green spaces into cities to enhance air quality, sanitation, and overall public health (Rosenzweig et al., 1992). As the value of urban green spaces became more widely recognized, the methods for mapping them also evolved. Before the 1970s, green space mapping was conducted through manual cartographic techniques, field surveys, and aerial photography methods that were time-consuming, spatially limited, and often imprecise due to their reliance on human interpretation (Tomlinson et al., 2011).

This was the beginning of the modern green space movement, wherein urban parks were introduced as “lungs of the city.” As urban planning matured into the early 20th century, attention shifted from scattered green areas to more integrated and equitable systems of green space. Ebenezer Howard's Garden City Movement (1902) proposed self-contained communities surrounded by green belts that harmonized the urban and rural environment. This model promoted city-wide planning of green infrastructure, highlighting the role of nature in shaping urban development, health, and social cohesion (Howard, 1902). Following World War II, modernist planning ideologies, particularly those advocated by Le Corbusier, gave rise to the "towers in the park" model. High-rise buildings were surrounded by open green areas, but these spaces were often vast, undefined, poorly maintained, and lacked human scale, leading to their underutilization and deterioration (Jacobs, 1961). Concurrently, the boom in suburban development fostered a new ideal: the private backyard as the primary green space for middle-class families. This suburban shift led to significant disinvestment in urban public parks, many of which fell into neglect and were perceived as unsafe or unwelcoming. The late 20th century witnessed a major conceptual shift in the understanding of green space. Influenced by the rise of the environmental movement and the growing awareness of urban environmental crises, green spaces began to be recognized not merely as decorative or recreational, but as critical infrastructure. The Millennium Ecosystem Assessment (2005) helped reframe green areas as providers of essential ecosystem services, including carbon sequestration, water purification, temperature regulation, and biodiversity support. The term "Green Infrastructure" emerged to describe these interconnected networks of green space that offer multiple environmental, economic, and social benefits. This approach acknowledges green areas as multifunctional systems not isolated patches of greenery, but components of broader environmental and urban resilience strategies (Benedict & McMahon, 2006; Wolch et al., 2014).

The mapping of green spaces has undergone a significant transformation over time, evolving from basic manual methods to sophisticated, technology-driven processes. Initially, before the 1970s, green space mapping relied heavily on manual cartographic techniques, field surveys, and aerial photography. These traditional methods were time-consuming, spatially limited, and lacked accuracy due to reliance on human interpretation (Tomlinson et al., 2011). The advent of remote sensing technology in the 1970s marked a breakthrough. With the launch of Landsat satellites, particularly Landsat 1 in 1972, it became possible to acquire synoptic, repetitive, and objective observations of the Earth's surface. Although early sensors like the Multispectral Scanner (MSS) and Thematic Mapper (TM) provided coarse spatial resolution, they enabled large-scale monitoring of vegetation and land cover changes (Jensen, 2005).

The integration of Geographic Information Systems (GIS) with remote sensing data in the 1990s allowed more precise mapping of urban green spaces. UGS maps are mostly generated using different classification methods. This may include supervised classification method and unsupervised classification method (Richards, 1993). Supervised classification requires the analyst to provide training data samples of pixels with known land cover types. This information is used by the classification algorithm to learn the characteristics (such as spectral values) of each land cover class. The algorithm then applies this knowledge to classify all other pixels in the image. One of the most used supervised methods is the Maximum Likelihood Classification (MLC). This technique assumes that the data for each class follow a normal distribution. It uses both the average values (mean vectors) and how spread out the data are (covariance matrices) to classify pixels. MLC performs well when classes are clearly separated and normally distributed, but its accuracy decreases if the training data are poor or if the classes do not follow a normal distribution, supervised classification methods widely used to extract UGS (Richards, 1993). On the other hand, unsupervised classification does not require any prior knowledge of land cover types. Instead, the algorithm automatically groups pixels into clusters based on their spectral similarity using statistical methods. This is useful when field data are not available. One popular method is the Migrating Means Clustering (MMC) algorithm, which starts by assigning pixels to random cluster centres. These centres are then adjusted to reduce the overall error, resulting in better grouping of similar pixels. The unsupervised classification methods like k-means and ISODATA are used for urban land cover mapping but the accuracy of UGS mapping is lower due to insufficient use of spectral or spatial information while supervised classification methods widely used to extract UGS (Richards, 1993).

Overall, the evolution of green space mapping technology has transitioned from static, labour-intensive methods to dynamic, data-rich approaches, enabling cities especially in the Global South to better understand, plan, and manage their green infrastructure.

2.2 Why Urban Green Space matters

Urban green spaces are essential components of a sustainable city, offering a wide range of ecological, environmental, social, and economic benefits. From an ecological standpoint, green spaces act as vital ecological corridors that help preserve biodiversity and promote urban resilience. They provide critical habitats for flora and fauna, enabling the conservation of native species even in highly modified urban landscapes (Tzoulas et al., 2007). Green spaces play a major role in regulating microclimates, reducing urban heat island (UHI) effects by shading surfaces, evapotranspiration, and reducing the albedo of urban areas (Gill et al., 2007; Bowler et al., 2010). They also contribute to air purification by trapping dust, absorbing pollutants, and sequestering atmospheric carbon dioxide, thus improving urban air quality and mitigating climate change (Nowak et al., 2006).

Hydrologically, green spaces improve urban water management by enhancing rainwater infiltration, reducing surface runoff, and minimizing flood risks (Weng, 2001). Strategically planned vegetation can also reduce energy use by cooling buildings in summer and providing wind breaks in winter (Akbari, 2002). From a social perspective, green spaces promote physical activity, mental well-being, and psychological restoration. Numerous studies have demonstrated that access to green environments is associated with lower levels of stress, anxiety, and depression, while fostering social cohesion and encouraging outdoor activities (Ulrich et al., 1991; Maas et al., 2006). Particularly in densely populated and economically disadvantaged neighbourhoods, the presence of green spaces supports health equity and environmental justice (Jennings & Bamkole, 2019). In economic terms, green infrastructure enhances property values, attracts investment, and reduces healthcare costs by improving public health. Well-maintained parks and urban forests are often linked with increased real estate demand and higher land values (Conway et al., 2010).

Globally, the importance of green space has been increasingly recognized in urban policy and planning. International frameworks such as the UN's *Sustainable Development Goal 11* emphasize the need to “make cities and human settlements inclusive, safe, resilient, and sustainable,” including through the provision of accessible green spaces (UN-Habitat, 2016). However, despite this recognition, rapid urbanization, infrastructure development, and land-use changes have led to a net decline in per capita green space availability in many cities, especially in the Global South (World Bank, 2020). According to the World Health Organization (WHO), cities should ideally have a minimum of 9 square meters of green space per person, yet many urban areas fall far short of this standard (WHO, 2010). In fact, the global average urban green space per capita has been declining, with urban land expansion outpacing green infrastructure development (Seto et al., 2012).

In South Asia, which hosts some of the most densely populated and rapidly urbanizing cities, the situation is even more critical. Cities like Delhi, Dhaka, and Kathmandu have witnessed significant reductions in green cover due to unplanned urban sprawl and increasing pressure on land resources (Pandey et al., 2020). A study by Kabir & Parolin (2012) in Dhaka reported that green space per capita had declined to below 1 square meter, far less than the recommended threshold. Urbanization in the region is often informal and lacks strategic land

use planning, resulting in encroachment on natural areas and the loss of parks, wetlands, and community forests. Although some cities have initiated urban greening programs, these are often piecemeal and not integrated into broader spatial planning frameworks (Gopal & Nagendra, 2014).

Nepal, particularly its urban centres, mirrors the broader South Asian pattern. The rate of urbanization in Nepal has accelerated rapidly over the past two decades, with the urban population increasing from 17% in 2011 to over 21% in 2021 (CBS, 2021). This rapid growth has often occurred without adequate green space planning, leading to fragmentation and reduction of green areas in cities like Kathmandu, Pokhara, and Birgunj. Studies indicate that Kathmandu Valley lost a significant portion of its agricultural and forested land between 1990 and 2020 due to the expansion of built-up areas (Paudel et al., 2022). In Birgunj Metropolitan City, land-use change analysis has shown a decline in vegetation cover, especially in wards undergoing industrial and residential expansion (Yadav et al., 2023). Moreover, the per capita green space in Nepalese cities remains below WHO standards, with limited policy implementation and poor maintenance of existing green infrastructure.

However, there are also positive signs of change. Government initiatives such as the National Urban Development Strategy (2017) and local efforts to promote urban forestry, roadside plantations, and public parks reflect a growing awareness of the value of green space. Community-based Forest management around urban fringes has also contributed to maintaining some level of green cover. Yet, these efforts need to be better integrated into urban planning frameworks and supported by spatial data tools like remote sensing and GIS for effective monitoring and management.

2.3 Remote Sensing and GIS for Green Space Mapping

Remote sensing (RS) and Geographic Information Systems (GIS) have emerged as indispensable tools in the monitoring, assessment, and planning of urban green spaces. The rapid growth of urban areas and the dynamic nature of land-use change necessitate spatially explicit, timely, and cost-effective approaches for green space mapping, and RS-GIS technologies fulfil these needs efficiently (Weng, 2012). Remote sensing provides synoptic, repetitive, and multi-spectral observations of the Earth's surface, enabling researchers to detect, classify, and quantify vegetation cover over time. Satellite imagery such as from Landsat, Sentinel, MODIS, and SPOT has been extensively used to monitor urban greenery and vegetation dynamics (Lu & Weng, 2007).

GIS, on the other hand, is used to integrate spatial data, analyse patterns, and generate thematic maps for visualization and decision-making. When coupled with remote sensing data, GIS facilitates a comprehensive analysis of spatial distribution, fragmentation, and temporal change of green spaces. Techniques such as supervised classification, normalized difference vegetation index (NDVI), and object-based image analysis (OBIA) are commonly applied to delineate and quantify green spaces (Jensen, 2015). NDVI, derived from red and

near-infrared bands, is particularly effective for assessing vegetation health, density, and biomass (Rouse et al., 1974; Tucker, 1979).

Furthermore, time-series analysis using RS data allows for the detection of trends in vegetation cover such as increase, degradation, or encroachment making it possible to evaluate the effectiveness of urban planning interventions. The integration of ancillary data such as elevation models, land-use maps, and demographic information in GIS further enhances spatial analysis, enabling the identification of green space deficits, access inequality, and ecosystem service potential (Zhou & Wang, 2011). In recent years, high-resolution satellite data and UAV (drone) imagery have enabled finer-scale mapping and monitoring of small green patches, street trees, and rooftop gardens that were previously difficult to detect using moderate-resolution sensors (Gómez et al., 2016). The relevance of RS and GIS in developing countries like Nepal is particularly significant, where urban expansion is often unregulated, and ground-based monitoring is limited due to logistical and financial constraints. Several studies in Nepal, including in Kathmandu Valley, Pokhara, and Birgunj, have successfully applied Landsat and Sentinel imagery along with GIS tools to map urban green cover, assess land-use change, and inform green space planning (Paudel et al., 2022; Yadav et al., 2023). These technologies not only enhance transparency and accuracy in environmental assessments but also support evidence-based policy-making and sustainable urban development.

2.4 Mapping UGS by European Copernicus services

The accurate mapping of urban green spaces (UGS) has become increasingly important in the context of rapid urbanization, climate change, and sustainable urban planning. The Copernicus Earth Observation Program, managed by the European Commission in collaboration with the European Space Agency (ESA), provides a suite of free, high-resolution satellite data that are particularly valuable for environmental monitoring, including green space assessment (European Union, 2015). Among the various Copernicus satellites, Sentinel-2A, launched in 2015, has emerged as a highly effective tool for mapping urban vegetation. It offers a spatial resolution of 10 to 20 meters across spectral bands, including red-edge and near-infrared (NIR) bands, which are particularly useful for assessing vegetation health and density (Drusch et al., 2012). Sentinel-2A's high revisit frequency of five days (when combined with Sentinel-2B) enables consistent monitoring of green cover over time, making it well-suited for urban dynamic studies.

Researchers have extensively utilized Sentinel-2 imagery in green space mapping due to its capability to capture vegetation indices such as the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Red-Edge NDVI (RENDVI). For example, Li et al. (2019) used Sentinel-2A to map urban green spaces in Beijing, demonstrating that its spatial and spectral resolution allows for the differentiation of dense vegetation, sparse greenery, and impervious surfaces with high accuracy. Similarly, BSI (Bare Soil Index) and NDVI derived from Sentinel-2 data were employed by Zhang et al. (2020) to analyse green

space fragmentation in Shanghai, highlighting the usefulness of multi-spectral data in urban landscape ecology.

The Urban Atlas, a component of the Copernicus Land Monitoring Service (CLMS), provides harmonized land use data for European cities and is built using Sentinel imagery. While this product is Europe-focused, the methods underpinning it particularly the use of Sentinel-2 for supervised land cover classification are applicable globally. In South Asia and developing regions, Sentinel-2 data are increasingly being adopted for cost-effective and scalable green space mapping (Khawaldar et al., 2020).

In the case of Nepal, although large-scale green space datasets are limited, Sentinel-2 imagery has proven effective in mapping land cover in urban areas such as Kathmandu and Pokhara. For instance, Gautam and Xu (2021) utilized Sentinel-2A to classify green spaces and urban heat islands in Kathmandu, revealing the inverse correlation between green cover and land surface temperature. Their study emphasized the potential of Sentinel-2-based vegetation indices in urban microclimate research.

The application of supervised classification techniques, such as the Maximum Likelihood Classification (MLC), on Sentinel-2 data has yielded high classification accuracy in distinguishing vegetation types in heterogeneous urban environments (Grekousis et al., 2015). The availability of red-edge bands in Sentinel-2 also enhances vegetation discrimination, making it possible to estimate vegetation health, canopy cover, and even biomass under urban settings (Delegido et al., 2011).

Overall, the Copernicus Sentinel-2A mission provides a powerful and accessible platform for urban green space mapping. Its high-resolution, multi-spectral capabilities, frequent revisit time, and open data policy make it ideal for long-term vegetation monitoring, spatial planning, and biomass estimation in rapidly urbanizing cities like Birgunj. By leveraging Sentinel-2A, this research integrates global best practices into the local context, enabling accurate, scalable, and up-to-date mapping of urban green space.

2.5 Accuracy Assessment

Accuracy assessment is a crucial step in evaluating the quality of classified maps derived from remote sensing data. It provides a quantitative basis for determining how well the classified outputs correspond to real-world conditions. While overall accuracy, the proportion of correctly classified pixels that is a commonly used metric, it does not account for the possibility of agreement occurring by chance. To overcome this limitation, the Kappa coefficient (κ) is widely employed in remote sensing studies as a more robust statistical measure of classification accuracy (Congalton, 1991).

The Kappa statistic measures the degree of agreement between classified data and reference data while adjusting for chance agreement. It ranges from -1 to +1, where values ≤ 0 suggest no agreement or worse than random, values between 0.41 and 0.60 indicate moderate

agreement, 0.61 to 0.80 substantial agreement, and values above 0.80 represent almost perfect agreement (Landis & Koch, 1977). Kappa is calculated from the confusion matrix, which compares the predicted classes from the classification process to actual ground truth data. This matrix provides additional metrics such as user's accuracy, producer's accuracy, and error of omission and commission, offering a multi-dimensional assessment of classification reliability (Foody, 2002).

Several studies have demonstrated the utility of Kappa in urban green space mapping across different geographic regions. For instance, Alavipanah et al. (2015) applied the Kappa statistic in Tehran, Iran, to assess the accuracy of urban vegetation maps derived from Landsat imagery. They reported a Kappa value of 0.81, indicating almost perfect agreement between classified data and ground truth, which supported further analysis of urban heat island mitigation by green space. Similarly, Yuan and Bauer (2007) used the Kappa coefficient in the Minneapolis–St. Paul metropolitan area to validate classified images of vegetation change over time. Their results, which included Kappa values above 0.75, confirmed the reliability of the temporal classification and underlined the impact of land development on urban green areas.

In South Asia, the application of the Kappa coefficient in green space studies is growing. In India, Chaudhary et al. (2018) used Kappa statistics in Delhi to assess land use classification, where the green cover was a primary focus. Their classification of urban vegetation using Sentinel-2 data yielded a Kappa value of 0.79, validating their analysis of declining green space in expanding peri-urban areas. Similarly, Rimal et al. (2019) in Kathmandu, Nepal, utilized Kappa statistics in a study assessing urban expansion and its impact on green infrastructure. Their study showed a Kappa value of 0.84, which supported the reliability of their land use classification and reinforced concerns over the shrinking proportion of urban greenery.

2.6 Above Ground Biomass Estimation

Biomass estimation is an essential component in understanding ecosystem productivity and ecological health. Biomass refers to the total mass of living biological material, while Above Ground Biomass (AGB) specifically includes all the living vegetation above the soil surface, such as stems, branches, leaves, and flowers. Accurate estimation of AGB is critical for evaluating carbon sequestration potential, assessing forest resources, and developing climate mitigation strategies (IPCC, 2006). Given the impracticality of destructive sampling, especially in protected or urban areas, researchers rely on non-destructive allometric models to estimate biomass using tree parameters such as diameter at breast height (DBH) and tree height (H) (Chave et al., 2005).

In the context of Nepal, one of the most widely used methods for biomass estimation was developed by Sharma and Pukkala (1990). They formulated species-specific and generalized allometric equations for a wide range of forest trees found across Nepal's ecological zones,

including the Terai, Siwalik, and mid-hills. The models are applicable to major native species like *Dalbergia sissoo*, and *Acacia catechu* and have since been widely applied for carbon stock estimation at local and national levels.

The utility of Sharma and Pukkala's models has been demonstrated in various studies. For instance, Paudel and Sah (2015) applied these equations to estimate the AGB of collaborative forests in the Terai region, reporting strong accuracy and ease of use. Similarly, DFRS (2015) employed these models for the *State of Nepal's Forests* report, confirming their value for national forest inventory efforts. Additionally, REDD+ pilot projects in Nepal have relied on Sharma and Pukkala's equations for baseline carbon assessments and monitoring changes over time (Rai et al., 2016). These applications confirm the reliability and national acceptance of these equations in biomass and carbon stock estimation.

In recent years, the applicability of these models has expanded into urban forest. Urban green areas, though fragmented and often under threat due to infrastructure development, contribute significantly to ecosystem services, including carbon storage, air purification, and microclimate regulation (Kabisch et al., 2017). Many urban trees are either native or naturalized species for which Sharma and Pukkala's models are directly applicable. For example, trees such as *Ficus religiosa*, *Azadirachta indica*, and *Mangifera indica* are commonly found in Nepalese urban environments and share similar growth forms with species used in the original model calibration (Bhattarai et al., 2017).

The major advantage of Sharma and Pukkala's equations is that they require only basic tree measurements (DBH and height), which can be collected using simple tools. This makes them especially practical for large-scale green space assessments in cities where destructive sampling is neither ethical nor feasible. Furthermore, when integrated with GIS and remote sensing tools, these models enable spatial estimation of AGB across urban landscapes (Yadav et al., 2023). However, some limitations exist, particularly in accounting for ornamental or exotic species not originally included in the models, and differences in growth form between urban and forest trees.

Despite these challenges, Sharma and Pukkala's equations remain the most suitable and widely accepted method for AGB estimation in Nepal, both in forested and urban contexts. Their scientific robustness, ease of application, and regional calibration make them indispensable tools for biomass monitoring and sustainable land use planning in Nepalese cities. In urban centres like Birgunj, where green space is under pressure, the application of such models is crucial for estimating biomass and informing urban forestry strategies aimed at enhancing ecological resilience and climate adaptation.

2.7 Green space in Nepal

In Nepal, literature is relatively limited but growing for mapping green space. (Sharma et al.,2018) conducted green space mapping in Pokhara using Landsat data, found 7.6% reduction of green space cover between 2000 and 2020 mainly due to urban sprawl.

(Adhikari et al., 2022) reported that urban greenery in Kathmandu valley decreased by 21% over two decades. A study (Shrestha et al., 2019) assessed green space coverage using Landsat 8 OLI image and object- oriented classification, findings indicated that approximately 86% of Butwal`s area comprises green spaces, translating to about 623 sq. M per capita. Comparative studies from cities like Kathmandu, Pokhara and Butwal show declining trends in urban green spaces due to unplanned urban expansion. Birgunj has undergone significant land use changes due to rapid population growth, cross-border trade and infrastructure development (Dube, 2019). Open spaces are increasingly recognized in Nepal`s urban policy framework after the 2015 earthquake, emphasizing their role in disaster management and urban resilience (Upadhyay et al., 2020).

Despite economic significance as Nepal`s "Commercial Capital", Birgunj lacks published studies on the spatial extent and ecological evaluation of its urban green space. Detailed assessments to quantify its green cover along, evaluate biomass determination is almost missing. Updated and detailed LULC map, uncontrolled and unplanned urban area expansion, lack of technical expertise and financial resources for regular monitoring and planning, limited public engagement and awareness and implementation of policies are the major hindrance for green space management.

Green Space Mapping is now recognized globally as a foundational element of sustainable urban planning. Cities around the world are leveraging geospatial technologies to monitor and manage greenery with measurable benefits to public health, climate resilience and biodiversity. Several studies exist for Kathmandu and Pokhara, there is inadequate geospatial studies on green space in Birgunj, despite its status as a key economic and industrial hub. Most planning documents focus on infrastructure and trade but ignore ecological and spatial planning. This research addresses this gap by providing a green space baseline assessment using current remote sensing methods.

Chapter Three: MATERIAL AND METHOD

3.1 Study area

This study was conducted in Birgunj Metropolitan city (BMC), during the period during April 2025. BMC in Parsa District in Madhes Pradesh in southern Nepal Study area was situated at 27°0′N latitude and 84°52′E longitude at an altitude of 90m above sea level. BMC covers an area of 132.07 sq. Km. It comprises 32 wards and includes a mixture of urban, semi-urban and peri-urban area. It lies 135 km south of the capital Kathmandu, attached in the north to Raxaul on the border of the Indian state of Bihar. As an entry point to Nepal from Patna, Birgunj is known as the “Gateway of Nepal”. It is also called the “Commercial Capital of Nepal". The total population of BMC is 272,382 with female 130,187 and male 142,195 is residing which is the second largest population of terai and fifth largest city in Nepal. The city experiences high summer temperature often exceeding 35°C in May and June, between 34°C to 38°C or occasionally above 40°C in April and June. In contrast, the winter temperature ranging between 8°C TO 12°C. Majority of its precipitation occurs during monsoon season which last from mid-June to late September. The annual rainfall ranges from about 1300mm to 2800mm with an average of 1800mm. This seasonal concentration of rainfall plays a vital role in vegetation growth and affects biomass accumulation in urban green spaces. The city hosts some notable green areas, such as Ghantaghar Park, Vishwa Hindu Parishad Park, and roadside plantations, which play a vital role in providing ecological services like carbon sequestration, microclimate regulation, and recreational spaces.

The selection of BMC as the study area for green space mapping and Above Ground Biomass (AGB) estimation is based on its rapid urbanization, ecological importance, and the pressing need for sustainable urban planning. Evaluating the spatial distribution of green spaces and estimating biomass within the city can contribute to better policy making in terms of climate

resilience and air quality management, and urban sustainability.

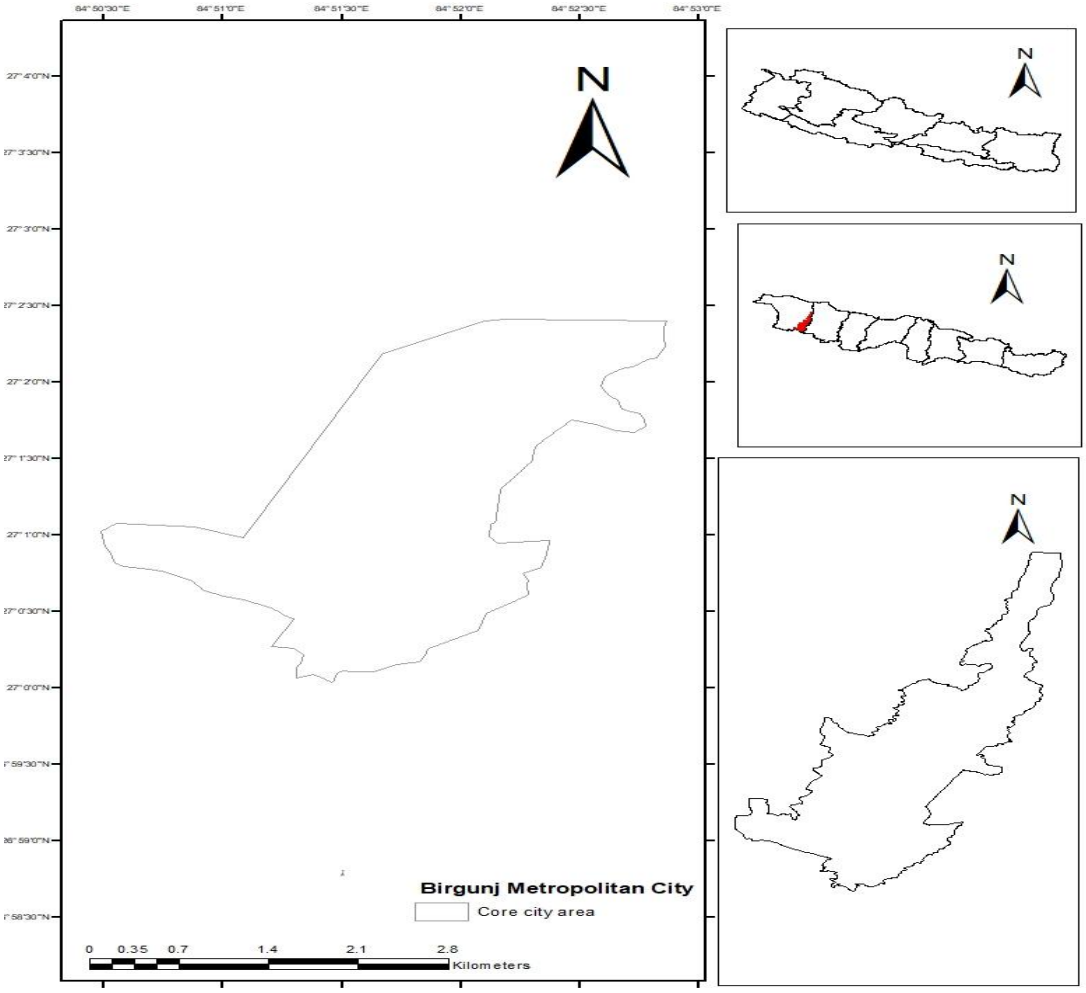


Figure 1: Study area map of Birgunj Metropolitan city, Parsa

3.2 Data collection and Processing

The data collection process for this study involved acquiring both remote sensing data and ground truth data to support accurate green space mapping and above ground biomass estimation. Figure 2 shows the overall flowchart of the study in which a high-resolution satellite image called sentinel-2A was collected from Copernicus Data Space Ecosystem. Then satellite image was further processed and classified using Arc GIS. Maximum likelihood supervised classification was employed to perform pixel based Supervised image classification categorizing the landscape into four major land use classes: agriculture, public ground, settlement, and water bodies. Within the formed categorization, sample points were generated through Create Accuracy Assessment Point in ArcGIS Spatial analysis tool as a result 302 sample points were selected. These points were navigated in the field using a handheld GPS. At each visited point, land cover type was verified and a plot of 50m * 50m was established, where tree species were identified and recorded, diameter at breast height

and total height of trees having DBH above 5cm were recorded for estimating AGB. Finally, data was statistically analysed and presented.

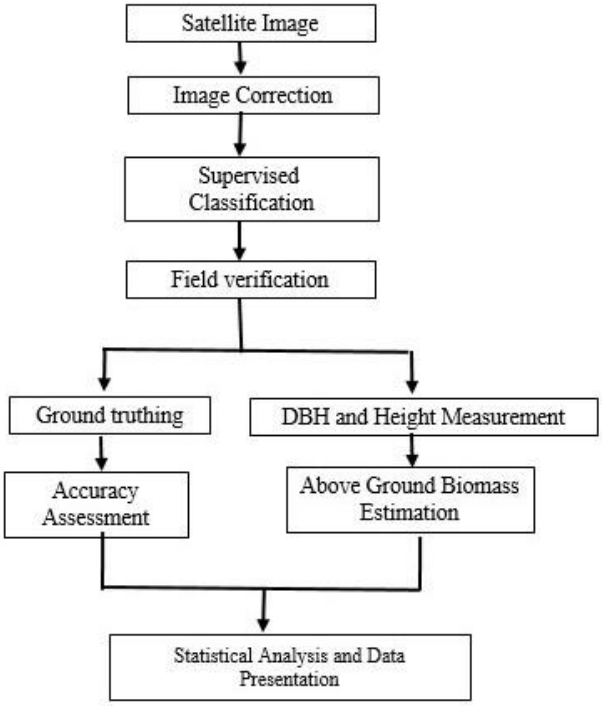


Figure 2: Flowchart of the study framework

3.2.1 Remote sensing data

For mapping green space, high resolution multispectral data provided by Copernicus Data Space Ecosystem was used(https://dataspace.copernicus.eu/). Sentinel-2A offers a high resolution that is 10-meter spatial resolution in different bands. Three visible and a near-infrared band with 10-meter resolution and six shortwave infrared bands with 20-meter resolution were used. The satellite image of 5th March 2025, was gathered to obtained cloud free data when cloud cover was less than 10%. Blue(B2), Green (B3), Red(B5) and Near infrared (B8) with wavelengths of 490nm, 560nm, 665nm and 843nm bands were used with 20 m × 20m resolution. The S2A data were processed and classified using ArcGIS, respectively All these bands were composited. Then shape file of BMC was added and then perform extract by mask to get desired area of interest. We preferred a supervised approach for higher accuracy. For each of the target land cover classes, a manually pre-classified training sample with a sufficient number of sample plots located evenly in the study area was created. The resulting set of totals 302 sample plots was fed into commonly used maximum likelihood classifier that was finally employed to perform Object –based Image classification of images. Geo-rectification was carried out and projected to GCS WGS 1984 UTM 45N (datum). Formed raster data was converted into vector. After that, vector image was dissolved according to green space classes.

3.2.2 Sampling Technique

First of all, purposive sampling technique was used to select sampling location based on the urban structure, land use structure and verification was done using Google earth pro.

A stratified random sampling technique was used to ensure adequate representation of different land cover types within the study areas. The following LULC classes were identified based on preliminary visual interpretation of sentinel-2A

- Agricultural land
- Settlement- house, buildings, hotels, grey infrastructure
- Ground- playground and bare ground both, parks
- Water bodies- river banks, ponds, lakes

For estimation of above ground biomass, sample plot designed for green space mapping was used because it covers different land cover types. So, there will be equal representation of all species found in different land cover.

3.2.3. Ground Truthing and field measurement

Ground truthing was done to check whether an object located in a map is present in real ground or not using GPS. Stratified Random Sampling with sampling intensity 0.1% on a proportional basis was used, choosing site which include water bodies, urban forest, agriculture land and public places such as park and ground. The entire study area of BMC was stratified into different land cover classes. Within formed strata, sample points were generated and plotted on the satellite image. A total of 302 sample plots of 50 × 50 sq. M size were selected, each layout in study area. These points were then navigated in the field using a hand-held GPS (Garmin etrex 10), ensuring representative coverage of different vegetation types and locations. At each visited point, the land cover type was verified, which were later used to train and validate supervised classification by constructing error matrix and data such as tree species, girth at breast height (GBH), and total height were recorded. Later on, GBH was converted into DBH. Height was recorded through rangefinder (Apresys Mini 1200). All individual species ≥ 5cm in DBH was recorded which falls under sample plot.

3.2.4. Biomass estimation

Non-destructive sampling method was used over destructive sampling because its efficient, environmentally friendly and sustainable method. Above Ground Biomass (AGB) estimation was carried out using species-specific allometric equations developed by Sharma and Pukkala (1990) for Nepal's tree species. These equations were chosen due to their regional relevance and suitability for subtropical and tropical urban forestry conditions. Field data on tree parameters such as Girth at Breast Height (GBH) and total tree height were collected from each sample plot established during ground truthing. The sample plots, each measuring 50 m × 50 m (2,500 m^2), were located in various types of urban green space. Stem biomass was determined using the equation 1.

Stem Biomass = Volume × Density.....(1)

where,

Density = Air-dried wood density in kg/m^3

Volume = Stem volume in m^3 (It was computed using the equation 2)

Stem volume

ln(v) = a + b ln(d) + c ln(h).....(2)

where,

ln = Natural logarithm to the base 2.71828.

v = Stem Volume (m^3) = exp [a + b × *ln* (DBH) + c × *ln* (h)]

d = DBH in cm

h = Total tree height in m

a, b and c are coefficients depending on species.

Note: Values were divided by 1000 to convert them to m^3 as the model gives stem volume in cubic centimetre.

Species-specific wood densities and value of a, b, c were obtained from FRA data analysis manual 2021 (Annex 1 and Annex 2).

3.3 Data analysis

Following the formulas of Moore and Chapman (1986), Shukla and Chandel (1980) and Dallmeier et al. (1992) quantitative structure parameters of investigated trees were calculated.

$$Frequency = \frac{Number\ of\ sample\ plots\ in\ which\ species\ occurred\ \times 100}{Total\ number\ of\ sample\ plots\ taken}$$

$$Relative\ Frequency(\ \%) = \frac{Frequency\ of\ individual\ species\ \times 100}{Total\ frequency\ of\ all\ species}$$

$$Density/ha = \frac{Total\ number\ of\ plants\ of\ any\ species\ \times 10000}{Total\ number\ of\ plots\ taken\ \times area\ of\ sample\ plot}$$

$$Relative\ Density\% = \frac{Density\ of\ individual\ species\ \times 100}{Total\ density\ of\ all\ species}$$

$$Basal\ area\ of\ a\ tree = \frac{(Diameter)^2 \times \pi}{4}$$

$$Relative\ Basal\ area\ \% = \frac{Basal\ area\ of\ all\ trees\ of\ species\ \times 100}{Total\ basal\ area\ of\ all\ species}$$

$$Coverage\ \% = \frac{Area\ occupied\ by\ species\ A\ \times 100}{Area\ of\ sample\ plot}$$

$$Relative\ coverage\ \% = \frac{Coverage\ of\ species\ A \times 100}{Total\ coverage\ of\ all\ species}$$

$$Impor\tan t\ \ Value\ Index = Relative\ Frequency + \ Relative\ Density + Relative\ Basal\ Area$$

3.3.1 Accuracy Assessment

Accuracy assessment of the classified satellite image was carried out to evaluate the reliability of the land cover classification, with a particular focus on green space mapping. Ground truth data collected through stratified random sampling were used as reference points. These points were overlaid on the classified output generated using supervised classification. A confusion matrix which is also termed as error matrix was developed by comparing the classified land cover categories with the corresponding ground truth data. From this matrix, standard accuracy metrics were calculated, including overall accuracy, user's accuracy, producer's accuracy, and the Kappa coefficient. The overall accuracy of the map was prepared and determined by computing Kappa coefficient by using the (equation 3)

$K= (P_o- P_e)/(1-P_e)$ .....(3)

where,

K = kappa coefficient

P_o= relative observed agreement among raters

P_e= hypothetical probability of chance agreement

Chapter Four: RESULT

4.1 Green space mapping and quantification

This study mapped out the urban green space within Birgunj Metropolitan City using GIS and remote sensing (RS) techniques. Figure 3 represents Agriculture, ground and waterbodies were classified as green space classes whereas settlement was classified as non-green space class. The result revealed that agriculture occupied the highest percentage of area with 53.06% (40.06 sq. Km) followed by settlement 20.53% (15.49 sq. km) then ground 18.37% (13.87 sq. Km) and least area by water bodies with 8.04% (6.07sq. Km) as shown in table 1. This presents a unique opportunity for proactive planning to integrate green networks within future development rather than retrofitting them later.

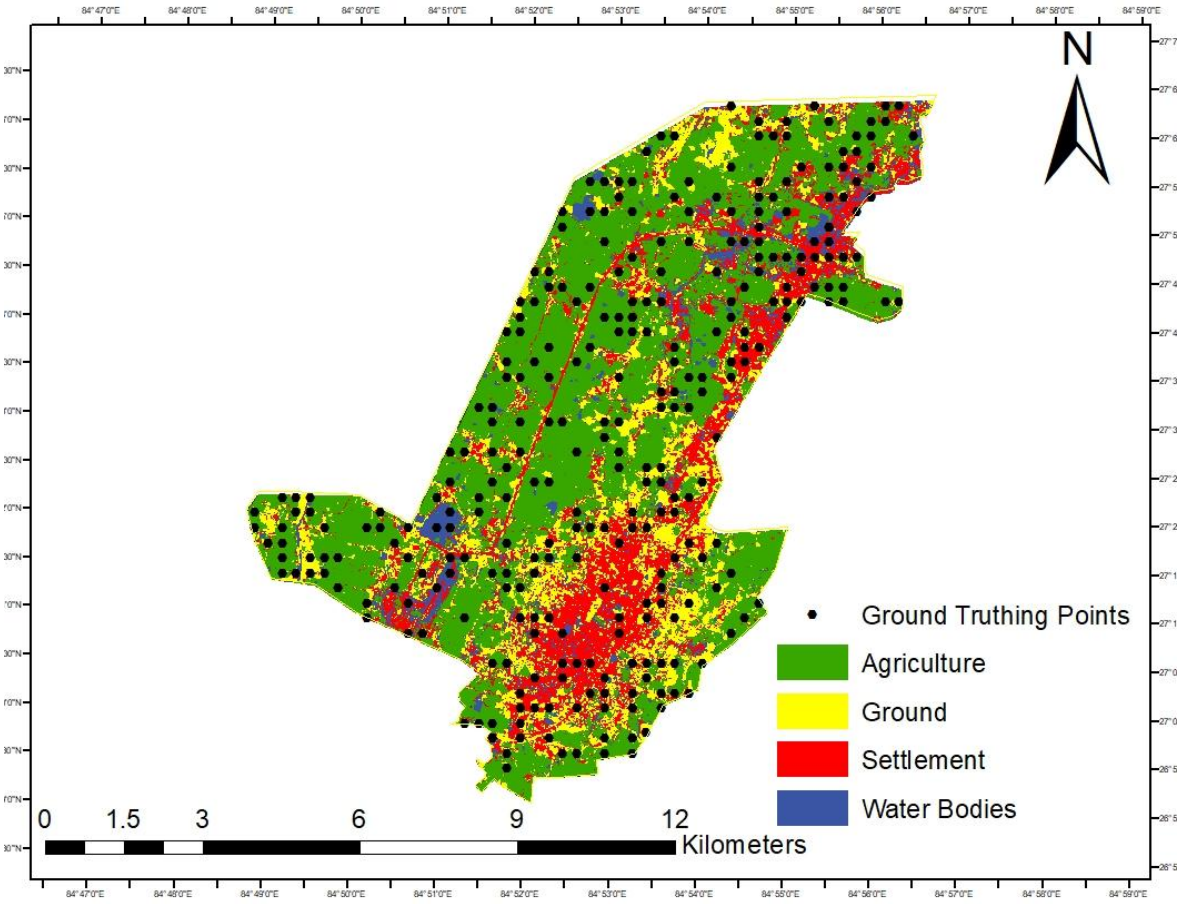


Figure 3: Spatial Distribution of green spaces in Birgunj Metropolitan city

Table 1: Land cover statistics of Birgunj Metropolitan

S.N.	Green space classes	Area (km sq.)	Percentage cover
1	Agriculture	40.06	53.06
2	Settlement	15.49	20.53
3	Ground	13.87	18.37
4	Water bodies	6.07	8.04
	Total	75.49	100

4.2 Accuracy assessment

The Kappa value of 0.871 was obtained which is an acceptable and strong level of agreement. Summary of the accuracy assessment is present in table 2 in the form of error matrix. The diagonal value for each city represents the correctly classified pixels which is 269. The ratio of their sum to the total number of tested pixels gives the total accuracy of the classification. User`s accuracy shows pixels incorrectly classified as a given class. Producer`s accuracy shows pixels of known class classified as other classes. The Kappa coefficient is an overall assessment of the accuracy of the classification which is 0.871 as shown in table 2.

Table 2: Calculation of accuracy and Kappa coefficient

	Agriculture	**Ground**	**Water bodies**	**Settlement**	**Users Accuracy**	**Producers Accuracy**	**Kappa coefficient**
Agriculture	98	3	2	4	0.915	0.859	0.871
Ground	7	84	2	2	0.884	0.943	
Water bodies	2	0	46	0	0.958	0.884	
Settlement	7	2	2	41	0.788	0.872	

4.3 Biomass and species composition

4.3.1 Species Composition

A total of twenty-three different species were recorded in study area with thirteen different families as shown in table 3. The species origin in the study area indicated that a total of 52% of the identified plant species were native to the region, whereas 48% were classified as exotic or non-native species as shown in figure (4). This near equilibrium between native and exotic species suggests that both local and introduced species contribute significantly to the existing vegetation structure and biodiversity within the urban green spaces of BMC.

Table 3: Species composition by family and origin

SN	Common name	Scientific name	Family	Species origin
1	Bell	*Aegle marmelos*	Rutaceae	Native
2	Rato siris	*Albizia chinensis*	Fabaceae	Native
3	Rukh Katar	*Artocarpus hetrophyllus*	Moraceae	Native
4	Neem	*Azadirachta indica*	Meliaceae	Exotic
5	Simal	*Bombax ceiba*	Malvaceae	Native
6	Rajbrikshya	*Cassia fistula*	Fabaceae	Native
7	Kapur	*Cinnamomum camphora*	Lauraceae	Exotic
8	Nariwol	*Cocos nucifera*	Arecaceae	Exotic
9	Siso	*Dalbergia sissoo*	Fabaceae	Native
10	Masala	*Eucalyptus supp.*	Myrtaceae	Exotic
11	Bar	*Ficus benghalensis*	Moraceae	Native
12	Kabro	*Ficus lacor*	Moraceae	Native
13	Peepal	*Ficus religiosa*	Moraceae	Native

14	Aap	*Mangifera indica*	Anacardiaceae	Native
15	Bakino	*Melia azadirach*	Meliaceae	Native
16	Moringa	*Moringa oleifera*	Moringaceae	Exotic
17	Kibhu	*Morus alba L.*	Moraceae	Native
18	Kadam	*Neolamarckia cadamba*	Rubiaceae	Native
19	Supari	*Phoenix sylvestris*	Arecaceae	Native
20	Ashoka	*Polyalthia longifolia*	Annonaceae	Exotic
21	Khair	*Senegalia catechu*	Fabaceae	Native
22	Jamun	*Syzygium cumimi*	Myrtaceae	Native
23	Teak	*Tectona grandis*	Lamiaceae	Exotic

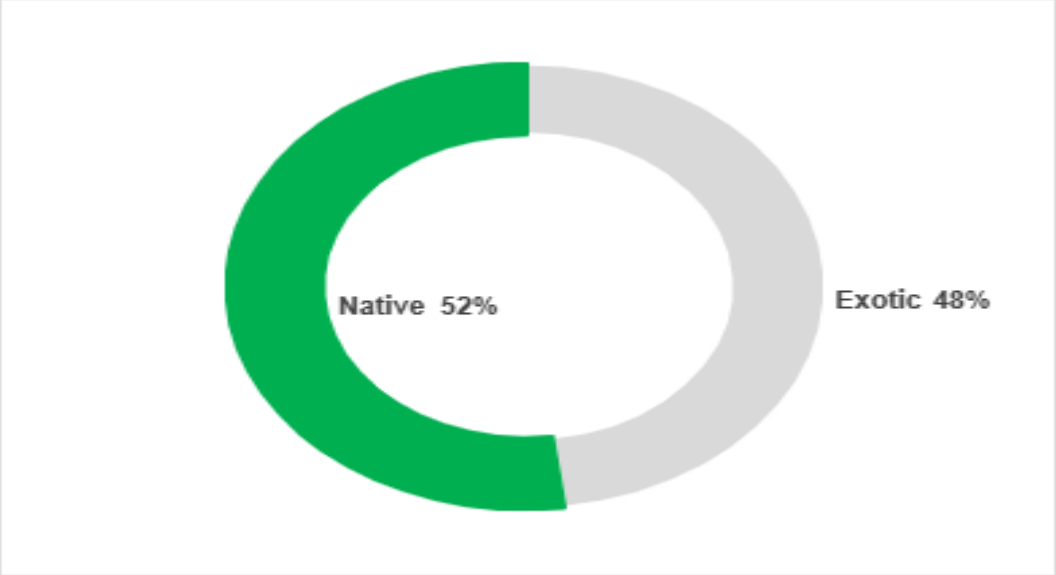


Figure 4: Composition of native and exotic species

4.3.2 Tree Attributes

A total of twenty-three different plant species were recorded in the study area among which *Ficus religiosa* (Peepal tree) has highest average diameter with 33.39 cm followed by *Ficus benghalensis* (Banyan tree) with 31.02 cm then by *Neolamarckia cadamba* (Kadam) with 29.45 cm. In contrast, the species with least average diameter were *Albizia chinensis* (Rato siris) with 5.1 cm. Species with tallest average height were *Eucalyptus supp.* (Masala) with 25.38 m followed by *Neolamarckia cadamba* (Kadam) with 23.07 m then by *Bombax ceiba* (Simal) with 21.87 m. In contrast, the species with shortest average height was *Cassia fistula* (Rajbrikshya) with 7.9m as shown in table 3. Similarly, the relative frequency%, relative density%, relative basal area%, Important Value Index (IVI) and relative coverage% of species were recorded. In the study area, it was observed that *Dalbergia sissoo* showed the maximum IVI (53.99) followed by *Mangifera indica* (41.56) and *Eucalyptus spp*. (39.07) as shown in figure 5. It was observed that *Neolamarckia cadamba* showed the maximum relative coverage (25.86%) followed by *Mangifera indica* (13.21%), *Dalbergia sissoo* (12.5%) and *Eucalyptus spp*. (10.86) as shown in figure 6.

Table 3: Average DBH, height, IVI and RC% present in BMC.

SN	Scientific Name	Avg. DBH	Avg. height	RF%	RD%	RB%	IVI	RC %
1	*Aegle marmelos*	14.8	12.3	0.625	0.32	0.62	1.57	0.26
2	*Albizia chinensis*	5.1	16	0.625	0.32	0.03	0.98	0.03
3	*Artocarpus hetrophyllus*	11.84	17.6	1.25	0.655	0.34	2.24	034
4	*Azadirachta indica*	9.05	13.43	8.75	4.91	2.62	14.0	2.62
5	*Bombax ceiba*	21.87	21.53	5	2.62	5.37	12.99	5.37
6	*Cassia fistula*	8.89	7.9	0.625	0.32	0.09	1.04	0.09
7	*Cinnamomum camphora*	6.4	15.75	1.25	0.65	0.1	2.0	0.1
8	*Cocos nucifera*	8.3	23.7	1.87	1.63	0.42	3.93	0.4
9	*Dalbergia sissoo*	12.03	15.9	23.12	18.36	12.51	53.99	12.5
10	*Eucalyptus supp.*	10.22	25.38	6.25	21.96	10.86	39.07	10.86
11	*Ficus benghalensis*	31.02	18.3	2.5	1.31	6.04	9.85	6.04
12	*Ficus lacor*	17.1	15.55	1.25	0.65	0.83	2.73	0.83
13	*Ficus religiosa*	33.39	18.43	4.37	1.96	10.5	16.84	10.5
14	*Mangifera indica*	18.5	18.14	16.87	11.47	13.21	41.56	13.21
15	*Melia azadirach*	19.29	19.05	2.5	1.31	1.98	5.79	1.98
16	*Moringa oleifera*	6.02	9.91	2.5	3.6	0.5	6.6	0.5
17	*Morus alba L.*	8.41	8.9	2.5	1.31	0.35	4.16	0.35
18	*Neolamarckia cadamba*	29.45	23.07	2.5	7.86	25.86	36.22	25.86
19	*Phoenix sylvestris*	6.85	18.5	1.25	0.98	0.17	2.4	0.17
20	*Polyalthia longifolia*	5.62	10.47	3.12	1.31	0.15	4.58	0.15
21	*Senegalia catechu*	10.8	15.5	2.5	1.63	0.94	5.07	0.94
22	*Syzygium cumimi*	26.57	21.5	1.25	0.65	2.3	4.2	2.3
23	*Tectona grandis*	8.88	21.02	7.5	14.09	4.44	26.03	4.44

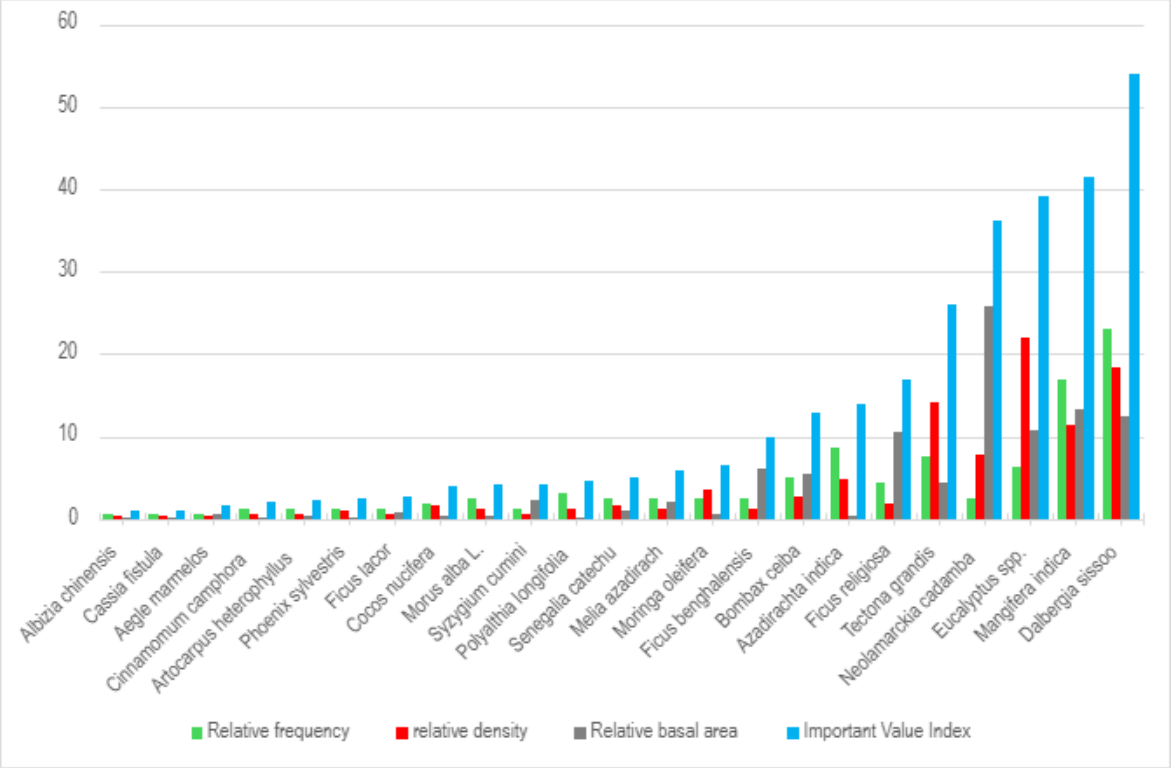


Figure 5: Important Value Index of species

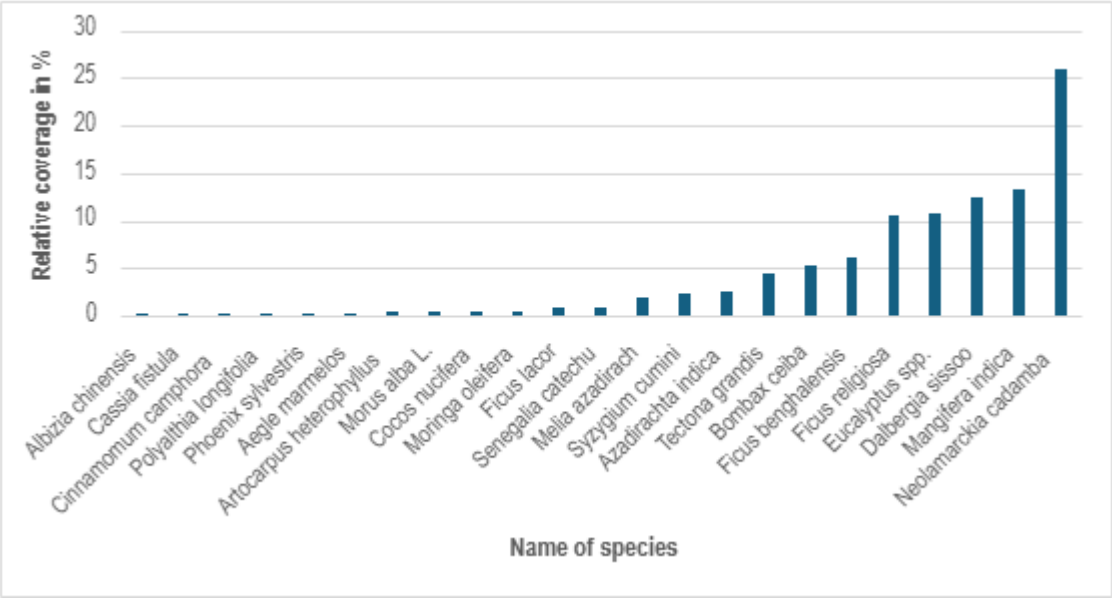


Figure 6: Relative coverage of species

4.3.3 Biomass Estimation

4.3.3.1 Biomass estimation by family

A total of thirteen family of different species were recorded in the study area. Among them Myrtaceae has the highest biomass with 153.956 kg/ha followed by the family Rubiaceae with 149.294 kg/ha then Fabaceae with 100.838 kg/ha. In contrast, family Rutaceae has the least biomass with 1.096 kg/ha as shown in table 4 and present in figure 7.

Table 4: Shows biomass of species by family

S N	Family	No of species	Biomass in kg/ha
1	Rutaceae	1	1.096
2	Malvaceae	1	4.450
3	Lauraceae	1	13.058
4	Arecaceae	2	17.793
5	Moraceae	5	18.053
6	Annonaceae	1	19.367
7	Moringaceae	1	42.869
8	Lamiaceae	1	47.569
9	Meliaceae	2	69.761
10	Anacardiaceae	1	81.784
11	Fabaceae	4	100.838
12	Rubiaceae	1	149.294
13	Myrtaceae	2	153.956
	Total	**23**	**719.894**

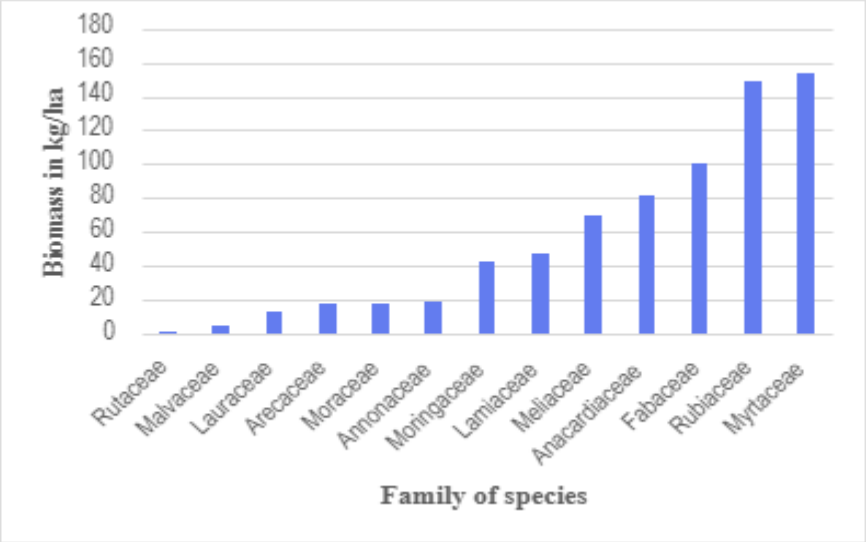


Figure 7: Above Ground Biomass by family

4.3.3.2 Biomass estimation by species origin

There were all together twenty-three species among which sixteen species were native and remaining seven species were exotic in BMC. The total average biomass of native species was 395.521 kg/ha, and exotic species was 324.6801 kg/ha as shown in figure 7.

Table 5: Biomass by species origin

SN	Common name	Scientific Name	Species origin	Average biomass in kg/ha
1	Bell	*Aegle marmelos*	Native	1.096
2	Rato siris	*Albizia chinensis*	Native	0.210
3	Rukh Katar	*Artocarpus hetrophyllus*	Native	1.166
4	Neem	*Azadirachta indica*	Exotic	44.506
5	Simal	*Bombax ceiba*	Native	4.450
6	Rajbrikshya	*Cassia fistula*	Native	5.736

7	Kapur	*Cinnamomum camphora*	Exotic	13.058
8	Nariwol	*Cocos nucifera*	Exotic	3.573
9	Siso	*Dalbergia sissoo*	Native	74.210
10	Masala	*Eucalyptus supp.*	Exotic	153.735
11	Bar	*Ficus benghalensis*	Native	7.945
12	Kabro	*Ficus lacor*	Native	5.32
13	Peepal	*Ficus religiosa*	Native	2.846
14	Aap	*Mangifera indica*	Native	81.784
15	Bakino	*Melia azadirach*	Native	25.255
16	Moringa	*Moringa oleifera*	Exotic	42.869
17	Kimbhu	*Morus alba L.*	Native	0.774
18	Kadam	*Neolamarckia cadamba*	Native	149.294
19	Supari	*Phoenix sylvestris*	Native	14.219
20	Ashoka	*Polyalthia longifolia*	Exotic	19.367
21	Khair	*Senegalia catechu*	Native	20.680
22	Jamun	*Syzygium cumimi*	Native	0.220
23	Teak	*Tectona grandis*	Exotic	47.569

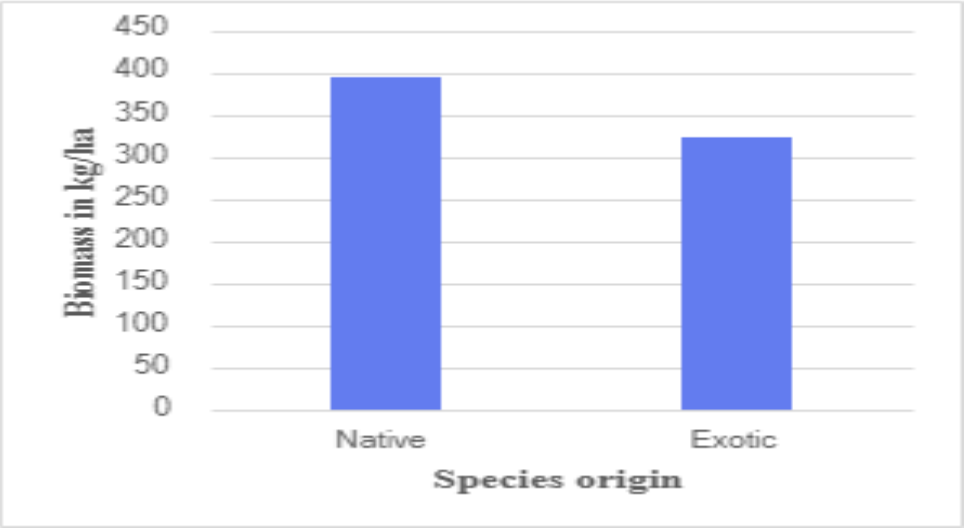


Figure 8: Biomass of species origin

4.3.3.3 Biomass estimation by species wise

Eucalyptus spp. (Masala) exhibited the highest average above ground biomass with 153.735 kg/ha, followed by *Neolamrckia cadamba* (Kadam) with 149.294 kg/ha then by *Mangifera indica* (Mango) with 81.784 kg/ha. In contrast, the species with least biomass was *Albizia chinensis* (Rato siris) with 0.210 kg/ha as shown in table 4 and represent by figure 7.

Table 6: Species identified in study area with their average biomass in kg/ha.

SN	Common name	Scientific Name	Family	No. Of trees	Average biomass in kg/ha
1	Bell	*Aegle marmelos*	Rutaceae	1	1.096
2	Rato siris	*Albizia chinensis*	Fabaceae	1	0.210

3	Rukh Katar	*Artocarpus hetrophyllus*	Moraceae	2	1.166
4	Neem	*Azadirachta indica*	Meliaceae	15	44.506
5	Simal	*Bombax ceiba*	Malvaceae	8	4.450
6	Rajbrikshya	*Cassia fistula*	Fabaceae	1	5.736
7	Kapur	*Cinnamomum camphora*	Lauraceae	2	13.058
8	Nariwol	*Cocos nucifera*	Arecaceae	5	3.573
9	Siso	*Dalbergia sissoo*	Fabaceae	56	74.210
10	Masala	*Eucalyptus supp.*	Myrtaceae	67	153.735
11	Bar	*Ficus benghalensis*	Moraceae	4	7.945
12	Kabro	*Ficus lacor*	Moraceae	2	5.32
13	Peepal	*Ficus religiosa*	Moraceae	6	2.846
14	Aap	*Mangifera indica*	Anacardiaceae	35	81.784
15	Bakino	*Melia azadirach*	Meliaceae	4	25.255
16	Moringa	*Moringa oleifera*	Moringaceae	11	42.869
17	Kimbhu	*Morus alba L.*	Moraceae	4	0.774
18	Kadam	*Neolamarckia cadamba*	Rubiaceae	24	149.294
19	Supari	*Phoenix sylvestris*	Arecaceae	3	14.219
20	Ashoka	*Polyalthia longifolia*	Annonaceae	4	19.367
21	Khair	*Senegalia catechu*	Fabaceae	5	20.680
22	Jamun	*Syzygium cumini*	Myrtaceae	2	0.220
23	Teak	*Tectona grandis*	Lamiaceae	43	47.569
				305	**719.894**

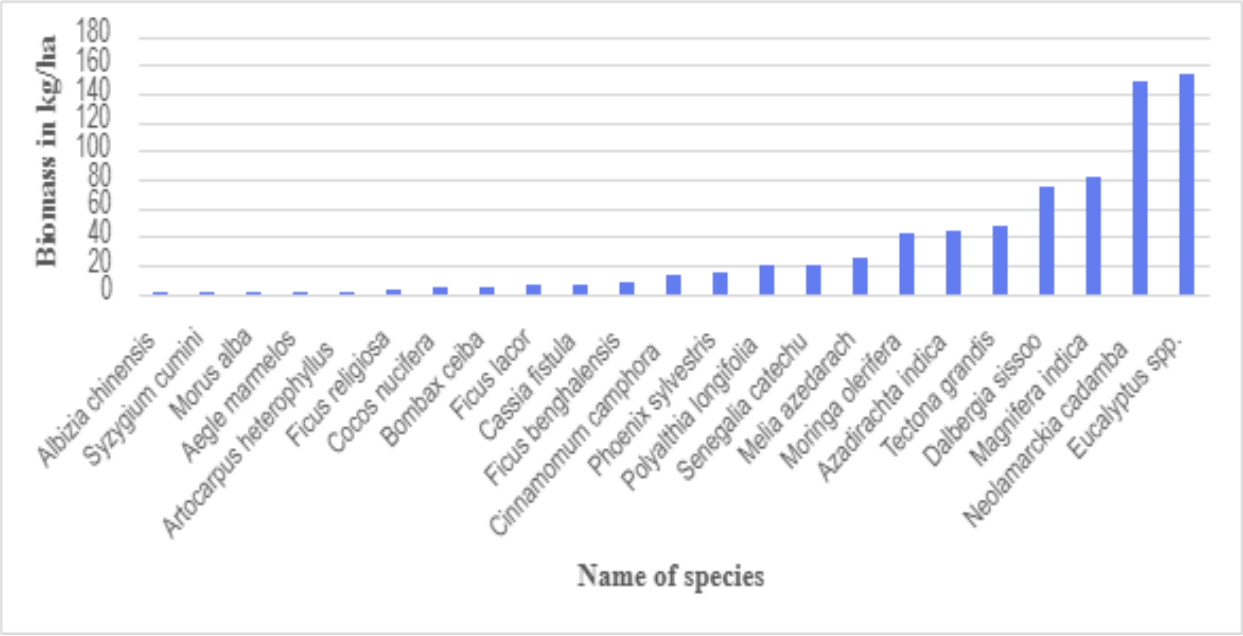


Figure 7: Above Ground Biomass by species

Chapter Five: DISCUSSION

5.1 Quantification of green space

The spatial analysis of land cover and above ground biomass (ABG) in Birgunj Metropolitan City (BMC) illustrates the transitional dynamics of a rapidly urbanization southern landscape of Nepal, wherein agricultural land persists as the dominant class but remains increasingly vulnerable to urban encroachment. The observation that cropland accounts for more than half of the total area (53.06%) underscores the continuing importance of peri-urban and sparsely populated wards in sustaining vital ecosystem services, including microclimate regulation, carbon sequestration, and groundwater recharge, a pattern similarly reported in Janakpur (Gautam & Mandal, 2020). Nonetheless, the growing proportion of built-up surfaces (20.53%), in conjunction with fragmented semi-natural elements such as water bodies (8.04%) and ground cover (18.37%), reflects an emergent imbalance between ecological integrity and urban expansion. The paucity of formally designated green infrastructure such as parks, tree-lined avenues, or urban forests; contracts with experience in Kathmandu Valley, where ward level planning supported by high resolution remote sensing has enabled targeted greening intervention (Paudel et al., 2022). This discrepancy highlights the absence of integrated spatial planning in BMC resulting in semi-urban green belts. In comparison with other cities, Birgunj shows a similar but distinct pattern of green space change. In Pokhara, green space loss is mainly linked to tourism-related development (Sharma et al., 2018), while in Kathmandu, vertical growth has encouraged the use of rooftop gardens as a way to adopt (Paduel et al., 2022). In contrast, Birgunj`s greenery is still mostly spread out across open ground, while makes it vulnerable to being replaced by new construction. This trend has serious consequences, as the loss of farmland and natural areas reduces the city`s ability to store carbon and disrupts the connections that plants and animals need to thrive. Thus, without deliberate policy interventions, BMC risks replicating regional patterns of green space fragmentation; however, the city simultaneously holds significant potential to leverage spatial governance mechanism to balance urban growth with ecological sustainability

5.2 Biomass estimation

A total of 23 different plant species from 13 families were recorded in sample plots of BMC. The results indicated that *Eucalyptus spp.* (Masala) stored the highest average biomass followed by *Neolamarckia cadamba* (Kadam tree) and *Mangifera indica* (Mango) respectively. The high aboveground biomass recorded in *Eucalyptus* spp. confirms the species' suitability for urban plantations focused on carbon sequestration and biomass accumulation. This aligns with previous studies in Nepal and other tropical regions, where Myrtaceae species consistently demonstrate high biomass storage potential (Ghimire et al., 2019; Dutta et al., 2020). However, reliance on a single genus can increase ecological vulnerability. Therefore, it is recommended to develop mixed-species plantation programs incorporating biomass-rich native species such as *Dalbergia sissoo* and *Mangifera indica* to enhance both carbon storage and ecosystem resilience.

When categorized by family, the Myrtaceae family exhibited the highest cumulative biomass followed by Rubiaceae and Fabaceae. Although Fabaceae ranked third in cumulative biomass, their ecological role is crucial for maintaining soil fertility and supporting other plant species (Singh et al., 2020). Including Fabaceae species in urban green space designs can enhance long-term biomass accumulation and improve soil nutrient status, contributing to sustainable urban forestry management.

5.3 Species origin

The analysis of species composition in Birgunj Metropolitan City revealed that native species accounted for 52%, while exotic species made up 48% of the total recorded species. This near balance between native and exotic species reflects both the persistence of local biodiversity and the significant influence of introduced species in the urban landscape. The predominance of native species is encouraging from an ecological perspective, as they are generally better adapted to the local climate, soils, and ecological interactions. Native species such as *Dalbergia sissoo*, *Neolamarckia cadamba*, and *Syzygium cumini* provide important ecosystem services, including habitat provision for native fauna, resilience against local pests and diseases, and contributions to cultural and religious values (CBD, 2010; Shrestha et al., 2020). Their presence supports the ecological integrity and functional stability of urban green spaces. In contrast, the relatively high proportion of exotic species such as *Tectona grandis*, *Eucalyptus species*, and *Polyalthia longifolia* highlights the influence of ornamental planting and afforestation initiatives aimed at aesthetic enhancement, rapid canopy cover, and specific functional purposes like shade or windbreaks (Kendal et al., 2014). While exotic species can offer valuable benefits, including fast growth and visual appeal, their ecological risks cannot be overlooked. Some exotic species can outcompete native species, alter soil properties, or become invasive under favourable conditions, potentially leading to reduced native biodiversity over time (Richardson et al., 2000).

A similar pattern was observed by Koirala & Das (2017) in Bharatpur, where ornamentals such as *Delonix regia* were widely planted alongside, native species like *Ficus religiosa* and *Ficus benghalensis*. In contrast, studies from Kathmandu Valley have reported a higher dominance of exotic species, with Pradhan & Shrestha (2020) finding over 60% of urban park trees to be exotic, primarily due to ornamental preferences and nursery availability. The proportion of native species in Birgunj aligns more closely with findings from Butwal, where Shrestha et al. (2019) recorded 55% native species, noting that municipal planting policies favoured indigenous trees for their ecological resilience. In Pokhara, however, Sharma et al. (2018) documented a gradual shift toward exotic plantings, driven by the tourism sector's demand for visually striking species such as *Delonix regia* and *Jacaranda mimosifolia*.

The assessment of ecological structure through parameters such as relative frequency, relative density, basal area, and IVI (Importance Value Index) highlighted that *Dalbergia sissoo* had the highest IVI (53.99), indicating it is the most ecologically dominant species in the sampled urban green areas. It was followed by *Mangifera indica* (41.56) and *Eucalyptus spp.* (39.07). High IVI suggests that these species are not only frequent and dense but also occupy a significant amount of the total basal area, reflecting their maturity and adaptation to the urban environment. Furthermore, *Neolamarckia cadamba* showed the highest relative coverage (25.86%), despite having a relatively moderate IVI. This suggests that while the species may not be the most frequently encountered or numerous, it covers a significant

canopy area, contributing to microclimate regulation and biomass. The dominance of certain species could be attributed to city level plantation schemes or residents' preferences. However, maintaining species diversity is essential to prevent monocultures and promote ecological balance. Mixed species plantations should be encouraged in future greening efforts.

The study results have significant implications for urban planning and green infrastructure development. The spatial imbalance of green space and limited representation of structured green areas suggest the urgent need to incorporate green belts, buffer zones, linear parks, and biodiversity corridors within urban land use plans. Additionally, enhancing tree biomass through proper species selection and plantation density can contribute to carbon sequestration, pollution reduction, and urban cooling, all of which are crucial for climate adaptation in lowland cities like Birgunj that experience extreme summer temperatures.

Chapter Six: CONCLUSION AND RECOMMENDATION

6.1 Conclusion

This study successfully conducted a comprehensive assessment of the urban green spaces in Birgunj Metropolitan City, providing a critical and timely ecological baseline for one of Nepal's most significant and rapidly urbanizing commercial centres. By integrating remote sensing techniques with field-based analysis, the research has yielded several key conclusions that are vital for the future of sustainable urban development in Birgunj. The classification results indicated that the largest land cover type was agriculture, covering 40.06 km² or 53.06% of the total study area. This dominance underscores the continued relevance of peri-urban and urban agriculture in BMC despite urban development pressures. It was followed by settlement areas (15.49 km², 20.53%), ground or open spaces (13.87 km², 18.37%), and water bodies (6.07 km², 8.04%). The inclusion of agriculture, ground, and waterbodies under green space classes reveals that 79.47% of BMC still comprises green or ecological zones. This provides a significant opportunity for the local government to incorporate green infrastructure into future urban planning, thereby reducing the need for expensive retrofitting of green areas. If managed properly, this green space network can serve as a buffer against pollution, urban heat islands, and biodiversity loss.

An overall Kappa coefficient of 0.871 was obtained from the classification accuracy assessment, indicating a high level of agreement between classified and reference data. A total of 302 samples were used for validation, of which 269 were correctly classified. This level of accuracy confirms the reliability of satellite-based remote sensing and supervised classification techniques in urban land cover mapping. The error matrix further highlighted the strengths and weaknesses of classification by showing individual User's Accuracy and Producer's Accuracy for each class. Such accuracy measures are essential to ensure the credibility of derived statistics, especially when they are used for biomass estimation or policy formulation. The successful accuracy assessment also reinforces the suitability of Sentinel-2A imagery for urban ecological studies in Nepal.

Biomass estimation revealed key insights into the carbon storage capacity of BMC's green cover. A total of 23 plant species from 13 botanical families were identified across the sampled plots. The Above Ground Biomass (AGB) of all 305 recorded trees was 3601.412 kg, signifying a moderate but valuable carbon sink within the urban setting. Among species, *Syzygium cumini* (Jamun) had the highest average AGB at 8.392 kg/ha, followed by *Ficus religiosa* (Peepal tree) with 7.532 kg/ha, and *Ficus benghalensis* (Banyan tree) with 7.580 kg/ha. These species not only possess large crown structures and wood density but also represent culturally and ecologically important components of Nepalese green spaces. In contrast, *Polyalthia longifolia* (False ashoka) exhibited the lowest biomass value at 0.169 kg/ha, possibly due to its smaller size and lower wood density.

When biomass was aggregated by botanical families, the Rubiaceae family stood out with the highest total biomass (7.058 kg/ha), followed by Moraceae (4.573 kg/ha) and Malvaceae (4.499 kg/ha). The dominance of Rubiaceae can be attributed to the presence of *Neolamarckia cadamba*, which showed a high individual AGB due to its large height and diameter. This finding is critical in species selection for future urban afforestation and carbon sequestration initiatives. It is important to note that not all families with high tree counts contributed proportionally to biomass. For example, Fabaceae, despite being one of the most frequent families, had lower average biomass compared to Rubiaceae and Moraceae. This reveals that species structure (DBH and height) plays a more vital role in biomass contribution than abundance.

Analysis of the species origin showed that 52% were native species, while 48% were exotic or non-native. This near equilibrium reflects a diverse urban flora, where both local and introduced species contribute to ecosystem services. Native species such as *Ficus religiosa*, *Dalbergia sissoo*, and *Mangifera indica* support indigenous biodiversity and cultural heritage. On the other hand, exotic species like *Tectona grandis, Eucalyptus spp*. and *Polyalthia longifolia* may have been planted for ornamental, shade, or rapid-growth purposes. However, a heavy reliance on exotic species in the long term may pose risks to ecosystem stability, including invasive behaviour, water competition, and lower support for native fauna. Therefore, a balanced planting strategy is recommended that prioritizes native species with high biomass potential and minimal ecological disruption.

The study's structural analysis showed that *Dalbergia sissoo* had the highest Importance Value Index (IVI) of 53.99, indicating its dominant presence in terms of frequency, density, and basal area. It was followed by *Mangifera indica* (41.56) and *Eucalyptus spp.* (39.07). These species play a crucial role in urban forest structure, offering shade, habitat, and biomass simultaneously. In terms of relative coverage, *Neolamarckia cadamba* had the maximum with 25.86%, a significant figure suggesting its spatial dominance. Structural metrics like IVI are valuable for prioritizing urban forest management and conservation, as they identify ecologically significant species for protection, propagation, or replacement.

The findings of this study offer valuable insights into the composition and biomass of urban green spaces in Birgunj Metropolitan City. With over half of the urban landscape still covered by agricultural and green spaces, there is significant potential for sustainable urban planning. Strategic management of dominant species, enhancement of native vegetation, and controlled use of exotics can improve carbon sequestration, urban resilience, and biodiversity conservation. This study demonstrates the usefulness of remote sensing combined with field validation in estimating green space dynamics and ecosystem services in urban settings. Future studies should incorporate temporal data to assess green space change over time and evaluate the effects of urban expansion on biomass and biodiversity.

6.2 Recommendation

In urban greening initiatives, native species should be promoted and prioritized in plantation programs as they are well-adapted to local conditions and play a key role in supporting biodiversity and ecosystem resilience. At the same time, biomass-rich species should be encouraged and enhanced to maximize ecological benefits such as carbon storage and climate regulation. To ensure long-term sustainability, species diversity must be maintained, as it helps reduce vulnerability to pests, diseases, and the impacts of climate change. Public engagement and awareness programs are equally important to foster community participation and stewardship in the protection and management of urban green spaces. Additionally, periodic monitoring using remote sensing and GIS should be conducted to track changes over time, providing essential data for informed planning, management, and conservation of urban ecosystems.

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ANNEXES

Annex 1: Equations for Volume and Bark Ratio

Genus	Species	Local name	definition	a	b	c	sf	Remarks
Acacia	*Acacia catechu*	Khair	NA	2.3256	1.6476	1.0552	0.12	Total volume
Dalbergia	*Dalbergia sissoo*	Sissoo	NA	2.1959	1.6567	0.9899	0.12	Total Volume
Class A	NA	NA	Miscellaneous in Terai	2.3993	1.7836	0.9546	0.16	Total volume

Annex 2: Density of species

Genus	Species	Local Name	Density
Acacia	*Acacia catechu*	Khair	960
Dalbergia	*Dalbergia sissoo*	Sissoo	780
Terai Miscellenous	Terai hill Miscellenous	Terai hill Miscellenous	674

PHOTO PLATES



Photo1: Land class classified as agriculture



Photo2: Data recording



Photo3: Degraded wetland with water hyacinth



Photo5: Sides of water bodies can be used as green space

BIOGRAPHICAL SKETCH



My name is Susmita Khadka. I was born on 25th of Jan 2001 to Janak Khadka and Sita Khadka in Dhading district of Nepal. I completed my SEE in the year 2074 B.S. and +2 in 2076 B.S. from Ed-Mark College. Currently, I am pursuing my bachelor's degree in forestry, under Forestry and Agriculture University (AFU), College of Natural Resource Management, Katari. I am passionate about urban forest and aspire to make a meaningful impact in the field of urban forestry.